

6-

```
postgres-#  
CREATE VIEW student_subjects_view AS  
  
SELECT  
    s.id AS student_id,  
    s.e_name AS student_name,  
    t.track_name,  
    sub.id AS subject_id,  
    sub.sub_name AS subject_name,  
    'Required' AS subject_type  
FROM  
    student s  
JOIN  
    track t ON s.track_id = t.id  
JOIN  
    track_sub ts ON t.id = ts.track_id  
JOIN  
    subject sub ON ts.sub_id = sub.id  
postgres-#
```

1-

```
postgres=# SELECT  
    s.sub_id AS subject_id,  
    s.sub_name AS subject_name,  
    s.max_score  
FROM  
    subject s  
WHERE  
    NOT EXISTS (  
        SELECT 1  
        FROM track_sub ts  
        WHERE ts.sub_id = sub_id  
    )  
ORDER BY  
    s.sub_name;  
subject_id | subject_name | max_score  
-----+-----+-----  
2 | Calculus I | 100  
1 | Database Systems | 100  
3 | Data Structures | 100  
(3 rows)  
  
postgres=#
```

2-

```
postgres=# SELECT
    e_name AS student_name,
    EXTRACT(YEAR FROM AGE(birth_date)) AS age
FROM
    student
ORDER BY
    e_name;
 student_name | age
-----+-----
    arwa      |  25
    aya       |  24
    menna     |  23
(3 rows)

postgres=#
```

3-

```
postgres=# SELECT
    s.e_name AS student_name,
    sub.sub_name AS subject_name,
    ROUND(CAST(g.grade AS numeric)) AS rounded_score,
    sub.max_score
FROM
    student s
JOIN
    grades g ON s.stu_id = g.stu_id
JOIN
    subject sub ON g.sub_id = sub.sub_id
ORDER BY
    s.e_name,
    sub.sub_name;
 student_name | subject_name | rounded_score | max_score
-----+-----+-----+-----
(0 rows)

postgres=#
```

4-

```
postgres=# SELECT
    e_name AS student_name,
    EXTRACT(YEAR FROM birth_date) AS birth_year
FROM
    student
ORDER BY
    student_name;
 student_name | birth_year 
-----+-----
    arwa      |        1999
    aya       |        2000
    menna     |        2001
(3 rows)

postgres=#
```

5-

```
INSERT 0 1
postgres=# INSERT INTO exam (exam_id, date)
VALUES (
    (SELECT COALESCE(MAX(exam_id), 0) + 1 FROM exam),
    NOW()
)
RETURNING date;
    date 
-----
2025-04-30
(1 row)

INSERT 0 1
postgres=#
```

6-

```
postgres=# SELECT
    s.e_name AS student_name,
    ROUND(AVG(g.grade::numeric), 2) AS average_grade,
    COUNT(*) AS number_of_exams
FROM
    grades g
JOIN
    student s ON g.stu_id = s.stu_id
WHERE
    s.stu_id = 1
GROUP BY
    s.e_name;
 student_name | average_grade | number_of_exams 
-----+-----+-----
(0 rows)

postgres=#
```

7-

```
postgres=# UPDATE student
SET email = REPLACE(LOWER(email), '@gmail.com', '@iti.com')
WHERE LOWER(email) LIKE '%@gmail.com';
UPDATE 3
postgres=#
```

8-

```
postgres=# SELECT
    exam_id ,
    date AS exam_date,
    CURRENT_DATE - date AS days_difference
FROM
    exam
ORDER BY
    date DESC;
 exam_id | exam_date | days_difference
-----+-----+-----
      4 | 2025-04-30 |              0
      5 | 2025-04-30 |              0
      3 | 2024-03-20 |             406
      2 | 2023-12-10 |             507
      1 | 2023-09-15 |             593
(5 rows)
```

9-

```
postgres=# SELECT
    stu_id,
    e_name AS student_name,
    email
FROM
    student
WHERE
    email NOT LIKE '%.com'
    OR email IS NULL
ORDER BY
    student_name;
 stu_id | student_name | email
-----+-----+-----
(0 rows)

postgres=#
```

10-

```
postgres=# SELECT
    exam_id,
    TO_CHAR(date, 'MM/DD/YYYY') AS formatted_date
FROM
    exam
ORDER BY
    date;
 exam_id | formatted_date
-----+-----
      1 | 09/15/2023
      2 | 12/10/2023
      3 | 03/20/2024
      4 | 04/30/2025
      5 | 04/30/2025
(5 rows)
```

```
postgres=#
```